

what is the VC dimension of: $(x + y + 1)^2$

Let's be very precise about **what hypothesis class you mean**, because that determines the VC dimension.

Given expression

$$(x + y + 1)^2$$

This is a **fixed function**, not a family with parameters.

Case 1 (most likely what your question means)

Hypothesis class

$$\mathcal{H} = \{h(x, y) = (x + y + 1)^2\}$$

There are **no free parameters**. The function is completely fixed.

VC dimension

$$\text{VCdim}(\mathcal{H}) = 0$$

Why?

- VC dimension measures the largest number of points that can be **shattered**
- Shattering means realizing **all possible labelings**
- With **only one fixed function**, you **cannot even flip a single label**
- Even **1 point** cannot be shattered (you only get one labeling)

✅ **Conclusion:**

$$\text{VCdim} = 0$$

Case 2 (common exam trick)

If instead the hypothesis class were:

$$\mathcal{H} = \{\text{sign}((x + y + b)^2) \mid b \in \mathbb{R}\}$$

Then note:

$$(x + y + b)^2 \geq 0 \quad \text{for all } x, y$$

So:

$$\text{sign}((x + y + b)^2) = +1 \quad \text{always}$$

Still only **one labeling**.

$\text{VCdim} = 0$

Case 3 (what people sometimes *think* it is, but it's not)

If you instead had a **general quadratic classifier**:

$$ax^2 + by^2 + cxy + dx + ey + f$$

Then:

- 6 parameters
- VC dimension in $\mathbb{R}^2$ is **6**

But **your function is NOT this** — it is one fixed quadratic curve.

Final answer (for your exact question)

$\text{VCdim}((x + y + 1)^2) = 0$

If you want, I can:

- Show a **1-point counterexample**
- Compare this to **linear vs quadratic VC dimension**
- Tie it directly to your earlier **VC = d+1 vs d+2 confusion**

Just tell me 👍



$$x^2 + y^2 + 2xy + x + y + 1$$

how is this VC = 0?

Great question — this is exactly the subtle point exams like to test.

Let's slow it down and be **100% explicit about the hypothesis class**, because **VC dimension is never about a single formula's algebraic complexity**.

Step 1: What you wrote

$$x^2 + y^2 + 2xy + x + y + 1$$

Yes — algebraically this **looks "rich"** (quadratic terms, cross term, etc.).

But VC dimension does **NOT** depend on how complicated the polynomial *looks*.

Step 2: What is the hypothesis class?

If your hypothesis class is:

$$\mathcal{H} = \{h(x, y) = x^2 + y^2 + 2xy + x + y + 1\}$$

then:

- There are **no parameters**
- No coefficients can change
- No threshold can change
- No sign flip can change

This is a **single function**

Step 3: VC dimension definition (key!)

VC dimension = maximum number of points that can be **shattered**, meaning:

For **every possible labeling** of those points, there exists **some hypothesis in the class** that realizes it.

Step 4: Try to shatter 1 point

Take **one point** (x_1, y_1) .

To shatter it, you must be able to label it:

- positive ✓
- negative ✓

But with **only one function**, the label is **fixed**.

Example:

$$h(0, 0) = 1 \Rightarrow \text{label is always } +$$

You **cannot** make it negative.

✗ Cannot shatter even 1 point.

Step 5: Therefore

$$\text{VCdim} = 0$$

⚠ **Important misconception (very common)**

✗ **WRONG intuition:**

"It's quadratic, so VC dimension must be high"

✓ **CORRECT rule:**

VC dimension depends on the number of free parameters, not the degree of the polynomial

Step 6: Contrast with a class that has $VC > 0$

If instead you had:

$$\mathcal{H} = \{\text{sign}(ax^2 + by^2 + cxy + dx + ey + f)\}$$

Now:

- 6 free parameters
- VC dimension = **6** in $\mathbb{R}^2$

This is a **completely different hypothesis class**.

Key takeaway (memorize this)

A single fixed function always has VC dimension **0**, no matter how nonlinear it looks.

One-line exam answer

Even though $x^2 + y^2 + 2xy + x + y + 1$ is a quadratic polynomial, it defines a single fixed hypothesis with no adjustable parameters, so it cannot shatter even one point. Therefore its VC dimension is **0**.

If you want, I can now:

- Give you a **visual counterexample**
- Compare this to **$VC = d+1$ for linear separators**
- Show how adding **just one parameter** changes VC from $0 \rightarrow 1$



↳ VC dimension = **6** in $\mathbb{R}^2$

isnt this supposed to be $6 + 1$?

Short answer: **No — it is not $6 + 1$** . The correct VC dimension is **6**, not 7, **and only when the coefficients are free parameters**.

Let's pin this down cleanly.

Where the "6 + 1" intuition comes from

People often memorize:

- **Linear separators in $\mathbb{R}^d$** have VC dimension $d + 1$

That rule comes from:

$$\text{sign}(w^\top x + b)$$

- d weights $w_1, \dots, w_d$
- **1 bias term b**

So total $d + 1$ parameters $\rightarrow \text{VCdim} = d + 1$

✅ This rule applies **only** to *linear* separators.

Now your quadratic case (the correct framing)

Consider the hypothesis class:

$$\mathcal{H} = \{\text{sign}(ax^2 + by^2 + cxy + dx + ey + f)\}$$

Count the parameters

Term	Parameter
x^2	a
y^2	b
xy	c
x	d
y	e
constant	f

➡ **6 free parameters total**

Why it is not 6 + 1

The constant term f **already is the bias**.

There is **no extra threshold** floating outside the sign:

$$\text{sign}(g(x, y)) \quad (\text{threshold fixed at } 0)$$

So:

-  not "6 coefficients + 1 bias"
-  the bias is **already included** as f

Key theoretical fact (this is the exam-ready line)

The VC dimension of polynomial threshold functions equals the number of independent parameters.

So here:

$$\text{VCdim} = 6$$

Why your original expression has VC = 0

Your original function:

$$x^2 + y^2 + 2xy + x + y + 1$$

has:

- **no adjustable parameters**
- **exactly one hypothesis**

Therefore:

$$\text{VCdim} = 0$$

Final summary table

Hypothesis class	VC dimension
Fixed polynomial (your exact expression)	0
General quadratic in $\mathbb{R}^2$	6
Linear separator in $\mathbb{R}^2$	3

One-sentence rule to remember

VC dimension = number of free parameters, not polynomial degree, and the bias term is already counted.

If you want, I can:

- Show **explicitly how 6 points can be shattered**
- Map this to **feature lifting** into $\mathbb{R}^6$
- Connect this to **VC = d+1 vs m+1 confusion**

Just say the word.



give me the rules on how to determine VC, depending on the separator

Perfect — here's a **clean, exam-ready rulebook** for determining **VC dimension based on the separator / hypothesis class**.

This is exactly the mental checklist you want during an ML theory exam.

VC DIMENSION — RULES & CHECKLIST

RULE 0 (ALWAYS CHECK THIS FIRST)

Are there free parameters?

- NO free parameters $\rightarrow \text{VC} = 0$
- YES $\rightarrow$ continue

VC dimension is about the *size of the hypothesis class*, not how complicated the formula looks.

RULE 1: Linear separators (hyperplanes)

Hypothesis class

$$\mathcal{H} = \{\text{sign}(w^\top x + b)\}, \quad x \in \mathbb{R}^d$$

VC dimension

$$\boxed{d + 1}$$

Why?

- d weights
- 1 bias
- Hyperplanes can shatter $d + 1$ points in general position

Examples

Space	VC dim
1D threshold	2
2D line	3
3D plane	4

RULE 2: Linear separators without bias

Hypothesis class

$$\mathcal{H} = \{\text{sign}(w^\top x)\}$$

VC dimension

$$\boxed{d}$$

Removing the bias removes **one degree of freedom**.

RULE 3: Polynomial threshold functions

General form

$$\text{sign} \left(\sum_i \theta_i \phi_i(x) \right)$$

VC dimension

number of independent features

Think: **linear separator in feature space**

Example: Quadratic in $\mathbb{R}^2$

Features:

$$\phi(x, y) = (x^2, y^2, xy, x, y, 1)$$

Number of features = 6

$$\boxed{\text{VC} = 6}$$

🚫 **No extra "+1"** — the constant term is already included.

RULE 4: Fixed polynomial / fixed function

Hypothesis class

$$\mathcal{H} = \{h(x)\}$$

VC dimension

0

No choice → no shattering

RULE 5: Axis-aligned separators**Example**

$$h(x) = \text{sign}(x_j - t)$$

VC dimension

1

For rectangles in $\mathbb{R}^d$:

 $2d$ **RULE 6: k-degree polynomial in $\mathbb{R}^d$**

Number of monomials:

$$\binom{d+k}{k}$$

VC dimension

$$\binom{d+k}{k}$$

Example

Quadratic ($k = 2$) in $\mathbb{R}^2$:

$$\binom{2+2}{2} = 6$$

RULE 7: Neural networks (high-level)

Network	VC dimension
Single perceptron	$d + 1$
MLP with W weights	$O(W \log W)$
Deep networks	Very large but finite

Exact VC often **not required on exams**, just scaling behavior.

RULE 8: Margin constraints (SVM intuition)

- **Hard margin linear SVM** → same VC as linear separator
- **Larger margin** → *effective* capacity smaller (but VC unchanged)
- VC $\neq$ generalization bound directly

EXAM DECISION CHECKLIST

Ask in order:

1. **Is this one fixed function?** → VC = 0
2. **Is it a linear separator?** → VC = $d + 1$
3. **Is it polynomial?** → count monomials
4. **Is bias already included?** → do NOT add +1
5. **Is it feature-lifted?** → VC = dimension of feature space

One-sentence summary (memorize this)

VC dimension equals the maximum number of points that can be shattered, which in practice equals the number of independent parameters (features), including the bias exactly once.

If you want next:

- **A visual table of separators vs VC**
- **Counterexamples** that kill common wrong answers

- A “when m+1 applies” proof sketch

Just say 👍



give me a table of all these for my exam

Here is a **clean, compact, exam-ready VC-dimension table** you can **memorize and reproduce**.

This covers **all separators** your exam is likely to test, with the correct rules and no traps.

VC DIMENSION — EXAM TABLE

Hypothesis class / separator	Hypothesis form	VC dimension	Key rule / intuition
Single fixed function	$h(x)$	0	No free parameters → cannot shatter even 1 point
1D threshold	$\text{sign}(x - t)$	2	Left / right of a point
Linear separator in $\mathbb{R}^d$	$\text{sign}(w^\top x + b)$	$d + 1$	d weights + bias
Linear separator (no bias)	$\text{sign}(w^\top x)$	d	Origin-anchored hyperplane
Line in $\mathbb{R}^2$	$\text{sign}(ax + by + c)$	3	Special case of linear
Plane in $\mathbb{R}^3$	$\text{sign}(ax + by + cz + d)$	4	Linear
Polynomial threshold	$\text{sign}(\sum \theta_i \phi_i(x))$	#features	Linear separator in feature space

Hypothesis class / separator	Hypothesis form	VC dimension	Key rule / intuition
Quadratic in $\mathbb{R}^2$	$ax^2 + by^2 + cxy + dx + ey + f$	6	6 monomials
Degree-k polynomial in $\mathbb{R}^d$	all monomials up to degree k	$\binom{d+k}{k}$	Count monomials
Axis-aligned threshold	$x_j > t$	1	One dimension
Axis-aligned rectangle ($\mathbb{R}^d$)	box classifier	$2d$	Min/max per dimension
k-NN (fixed k)	nearest neighbors	∞	Can shatter arbitrarily many points
Decision stump	one split	1	One threshold
Decision tree (depth k)	hierarchical splits	grows exponentially	Capacity grows fast
Single perceptron (NN)	linear neuron	$d + 1$	Same as linear
Neural net with W weights	general	$O(W \log W)$	Upper bound
Hard-margin SVM	max-margin hyperplane	$d + 1$	Margin $\neq$ VC
Fixed quadratic (no params)	your earlier example	0	Still one hypothesis



COMMON EXAM TRAPS (MEMORIZE)

✗ Trap 1: "Polynomial degree $\Rightarrow$ VC"

Wrong.

VC depends on **parameters**, not degree.

✗ Trap 2: "Add +1 for threshold"

Wrong if constant term already exists.

$$ax^2 + by^2 + \dots + f \quad \text{already includes bias}$$

✗ Trap 3: "Margin changes VC"

Wrong.

Margins affect **generalization bounds**, not VC dimension.

ONE-LINE RULE TO WRITE ON EXAM

VC dimension equals the maximum number of points that can be shattered, which in practice equals the number of independent parameters (features), counting the bias exactly once.

If you want next, I can:

- Compress this into a **half-page handwritten cheat sheet**
- Give **1-line justifications** for each row
- Show **visual shattering examples** for linear vs quadratic

Just tell me.



make the table shorter - use only the most important and general cases

Perfect — here is a **short, high-yield VC-dimension table** with **only the cases that actually show up on exams**.

This fits easily on **one notebook page** and covers ~90% of questions.

VC DIMENSION — CORE EXAM CASES

Separator / hypothesis class	Form	VC dimension	Rule to remember
Fixed function	$h(x)$	0	No parameters → no shattering
1D threshold	$\text{sign}(x - t)$	2	Left / right split
Linear separator in $\mathbb{R}^d$	$\text{sign}(w^\top x + b)$	$d + 1$	Weights + bias
Linear separator (no bias)	$\text{sign}(w^\top x)$	d	Origin-anchored
Polynomial threshold	$\text{sign}(\sum \theta_i \phi_i(x))$	#features	Linear in feature space
Quadratic in $\mathbb{R}^2$	$ax^2 + by^2 + cxy + dx + ey + f$	6	Count monomials
Degree-k polynomial in $\mathbb{R}^d$	all monomials $\leq k$	$\binom{d+k}{k}$	Feature count
Neural net (general)	W weights	$O(W \log W)$	Capacity scales with weights

EXAM CHECKLIST (1–2 seconds)

1. Is it one fixed function? → VC = 0
2. Is it linear? → VC = $d + 1$
3. Is it polynomial? → count monomials
4. Is bias already inside? → don't add +1
5. Is it a NN? → scale with weights

ONE-LINE MEMORY RULE

VC dimension = number of independent parameters (features), counting the bias exactly once.

If you want, I can:

- Turn this into a **handwritten-style cheat sheet**
- Add **tiny diagrams** for intuition
- Give **exam-style yes/no VC questions**

Just say 👍

